Grade

Tecnologico de Monterrey
Partial Exam 2 A
2026-Jan-May

Course: Calculus I | **ID-number:** ______________
Professor: Fco. Javier Vazquez Tavares | **Date:** 24/03/2026
Last Names and Names: ________________________________ | **Group:** 201

"*In accordance with Tec de Monterrey Student Cod of Honor, in this exam I agree to conduct myself guided by academic honestly and not to use unauthorized or illegal means to do so.*"

Signature: ______________

I Multiple option section. Read the following questions and identify the option that best answers each, then write ts letter on the line.

1. (4 **points**) _____ Identify the quotient rule for derivatives.

   (a) $\left(\frac{u}{v}\right)' = \frac{uv' - u'v}{v^2}$ (b) $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$ (c) $\left(\frac{m}{n}\right)' = \frac{mn' - n'm}{n^2}$ (d) $\left(\frac{m}{n}\right)' = \frac{n'm - m'n}{n^2}$.

2. (4 **points**) _____ Identify the product rule for derivatives

   (a) $(uv)' = u'v + uv'$ (b) $(uv)' = u'v + u'v$ (c) $(uv)' = uv' + uv'$ (d) $(uv)' = u' + v'$

3. (4 **points**) _____ Identify the chain rule for derivatives

   (a) $[u(v(x))]' = v'(u(x))$ (b) $[u(v(x))]' = u'(v(x))$ (c) $[u(v(x))]' = u'(v(x))v'(x)$ (d) $[u(v(x))]' = v'(u(x))u'(x)$

4. (4 **points**) _____ Which of the following derivatives is correct

   (a) $\frac{\mathrm{d}}{\mathrm{d}x}n^x = xn^x$ (b) $\frac{\mathrm{d}}{\mathrm{d}x}n^x = n^x\ln(n)$ (c) $\frac{\mathrm{d}}{\mathrm{d}x}n^x = ne^{x-1}$ (d) $\frac{\mathrm{d}}{\mathrm{d}x}n^x = \frac{1}{x}n^x$

5. (4 **points**) _____ Which of the following derivatives is correct

   (a) $\frac{\mathrm{d}}{\mathrm{d}x}\cos(x) = \cos(x)$ (b) $\frac{\mathrm{d}}{\mathrm{d}x}\cos(x) = -\sin(x)$ (c) $\frac{\mathrm{d}}{\mathrm{d}x}\cos(x) = \sin(x)$ (d) $\frac{\mathrm{d}}{\mathrm{d}x}\cos(x) = -\cos(x)$

6. (4 **points**) _____ Which of the following derivatives is correct

   (a) $\frac{\mathrm{d}}{\mathrm{d}x}\sin(x) = \cos(x)$ (b) $\frac{\mathrm{d}}{\mathrm{d}x}\sin(x) = -\sin(x)$ (c) $\frac{\mathrm{d}}{\mathrm{d}x}\sin(x) = \sin(x)$ (d) $\frac{\mathrm{d}}{\mathrm{d}x}\sin(x) = -\cos(x)$

7. (4 **points**) _____ Which of the following derivatives is correct

   (a) $\frac{\mathrm{d}}{\mathrm{d}x}\log_n(x) = \frac{1}{\ln(x)}$ (b) $\frac{\mathrm{d}}{\mathrm{d}x}\log_n(x) = \frac{1}{x\ln(n)}$ (c) $\frac{\mathrm{d}}{\mathrm{d}x}\log_n(x) = \frac{\ln(x)}{x}$ (d) $\frac{\mathrm{d}}{\mathrm{d}x}\log_n(x) = \frac{1}{x\ln(x)}$

II Derivatives by rules. Use the rules of derivatives that can be usefull to compute the derivatives of the following functions.

1. (10 **points**) Compute the derivative of the function $f(x) = e^{-x^2} \sin^2(x)$.

2. (10 **points**) Compute the derivative of the function $g(x) = \log\left[xe^{-x}\right]$.

III Critical points. In this section you task is to find <u>all</u> critical points of the functions in the given context.

1. (13 **points**) An open box is made from a 5×5 cm sheet by cutting squares of side x from the corner. The volume of the box is given by $V(x) = x(5-2x)^2$. Find the length of the squares that achieves the greater volume.

2. (13 **points**) The displacement of the planets over time can be modeled with the following function, $s(t) = 10\sin\left(\frac{\pi}{5}t\right) + 5\cos\left(\frac{\pi}{5}t\right)$. Find the time at which the planets are in a maximum and minimum displacements.

3. (13 **points**) The concentration of a drug in blood after t hours is given by $C(t) = te^{-kt}$. Find the time that gives the maximum concentration in terms of k.

4. (13 **points**) A rectangular plot with one side along a river needs fencing on three sides. If the area is fixed A and the cost is given by $C(x) = px + 2p\frac{A}{x}$, where x is the side along the river. Compute side length of the side of the river that minimize the fencing cost given that $A = 22500$ and $p = 100$.